

Cell-Free ISAC 系统鲁棒波束成形优化： 从非凸问题到可解 SDP

研究阶段报告

June 27, 2026

Abstract

本文针对无蜂窝集成感知与通信 (Cell-Free Integrated Sensing and Communication, ISAC) 系统, 研究在信道状态信息 (CSI) 存在估计误差的条件下, 联合波束成形优化的鲁棒设计问题。考虑通信用户服务质量 (QoS) 约束、感知检测概率约束、跟踪精度约束 (以 PCRB 表征) 以及接入点 (AP) 选择约束, 建立了以最小化系统总发射功率为目标的混合整数非线性规划 (MINLP) 问题。通过引入协方差矩阵变量、半定松弛 (SDR) 以及 S-Procedure, 将原始极度非凸问题严格转化为标准凸半定规划 (SDP), 并证明转化过程的等价性。进一步给出高斯随机化秩一恢复算法, 保证凸最优解可转化为物理可实现的波束向量。本文建立了从物理建模到凸优化求解的完整数学框架, 为后续算法实现与系统部署奠定理论基础。

Contents

1	Introduction	2
2	System Model and Problem Formulation	2
2.1	Cell-Free ISAC System Model	2
2.1.1	Communication Signal Model	3
2.1.2	Sensing Signal Model	3
2.1.3	Imperfect CSI Model	3
2.2	Problem Formulation	4
3	Problem Convexification and SDP Reformulation	4
3.1	Non-Convexity Analysis	4
3.2	Convexification Procedure	5
3.3	Final Convex SDP Problem (P3)	6
4	Beam Recovery and Algorithm Design	7
4.1	Rank-One Recovery via Gaussian Randomization	7
4.2	Computational Complexity Analysis	9
4.2.1	SDP Solver Complexity	9
4.2.2	Closed-Form Baseline Complexity	9
5	Conclusion	9

1 Introduction

集成感知与通信 (Integrated Sensing and Communication, ISAC) 作为第六代移动通信 (6G) 的关键使能技术, 通过在共享频谱和硬件资源上同时实现通信与雷达感知功能, 显著提升了频谱效率和系统能效 [?]. 与传统分离式设计相比, ISAC 通过波形共享和联合波束成形设计, 在通信性能与感知精度之间取得最优权衡, 已成为学术界和工业界的研究热点 [?].

在 ISAC 系统架构中, 无蜂窝 (Cell-Free) 网络通过分布式部署大量低功率接入点 (Access Point, AP), 由中央处理单元 (Central Processing Unit, CPU) 通过前传链路进行联合信号处理, 有效消除小区间干扰并提升覆盖均匀性 [?]. 将 Cell-Free 架构与 ISAC 相结合, 既能利用分布式 AP 的空间分集增益提升通信可靠性, 又能通过多视角观测增强目标感知与跟踪能力, 代表了未来无线网络的重要演进方向 [?].

然而, Cell-Free ISAC 系统的联合波束成形优化面临多重挑战。首先, 通信波束与感知波形在功率、空间方向图和时频资源上存在耦合, 导致优化问题高度非凸。其次, 实际系统中信道状态信息 (Channel State Information, CSI) 通过有限导频估计获得, 存在有界或随机的估计误差, 需要在优化设计中显式考虑鲁棒性。第三, 感知性能通常以 Cramér-Rao 下界 (CRB) 或预测 CRB (PCRB) 表征, 其关于波束变量的非线性依赖进一步增加了问题的复杂性。最后, AP 选择作为二进制决策变量, 使问题成为混合整数非线性规划 (Mixed-Integer Nonlinear Programming, MINLP), 在计算上属于 NP-hard 类。

针对上述挑战, 本文建立了 Cell-Free ISAC 系统鲁棒波束成形的完整数学框架, 核心贡献如下:

- 问题建模: 建立了包含通信 QoS、感知检测、跟踪精度和 AP 选择约束的标准 MINLP 问题, 显式刻画 CSI 估计误差对通信和感知性能的影响。
- 等价凸化: 通过协方差矩阵提升、半定松弛 (SDR) 和 S-Procedure, 将原始非凸问题严格转化为标准凸 SDP, 并证明转化过程的等价性。
- 求解算法: 给出基于高斯随机化的秩一恢复算法, 保证凸最优解可转化为物理可实现的波束向量, 同时提供功率缩放兜底方案。
- 复杂度分析: 对比 SDP 内点法求解与 ZF 闭式基准的复杂度, 为工程实现提供计算可行性评估。

Notation: Scalars are denoted by italic letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. $\mathbb{C}^{N \times M}$ and $\mathbb{H}^N$ denote the set of $N \times M$ complex matrices and $N \times N$ Hermitian matrices, respectively. $(\cdot)^T$, $(\cdot)^H$, and $\text{tr}(\cdot)$ denote transpose, conjugate transpose, and trace, respectively. $\mathbf{X} \succeq \mathbf{0}$ means $\mathbf{X}$ is positive semidefinite. $\|\cdot\|_2$ denotes the Euclidean norm.

2 System Model and Problem Formulation

2.1 Cell-Free ISAC System Model

考虑一个 Cell-Free ISAC 系统, 包含 M 个分布式 AP、 K 个单天线通信用户 (Communication User, CU) 和 P 个感知目标。每个 AP 配备 N_t 根发射天线, 通过理想前传链路与中央 CPU 连接。系统采用时分双工 (TDD) 模式, 利用信道互易性获取下行 CSI。

2.1.1 Communication Signal Model

用户 k 的接收信号可表示为:

$$y_k = \underbrace{\sum_{m=1}^M \mathbf{h}_{m,k}^H \mathbf{w}_{m,k} s_k}_{\text{desired signal}} + \underbrace{\sum_{j \neq k} \sum_{m=1}^M \mathbf{h}_{m,k}^H \mathbf{w}_{m,j} s_j}_{\text{multi-user interference}} + n_k \quad (1)$$

其中 $s_k \sim \mathcal{CN}(0, 1)$ 为发送符号, $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信波束成形向量, $\mathbf{h}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信信道, $n_k \sim \mathcal{CN}(0, \sigma_c^2)$ 为接收噪声。

为便于分析, 引入堆叠向量形式。定义全局通信信道 $\mathbf{h}_k \in \mathbb{C}^{MN_t}$ 和全局通信波束 $\mathbf{w}_k \in \mathbb{C}^{MN_t}$:

$$\mathbf{h}_k = [\mathbf{h}_{1,k}^T, \mathbf{h}_{2,k}^T, \dots, \mathbf{h}_{M,k}^T]^T, \quad \mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \dots, \mathbf{w}_{M,k}^T]^T \quad (2)$$

则接收信号可紧凑表示为:

$$y_k = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{j \neq k} \mathbf{h}_k^H \mathbf{w}_j s_j + n_k \quad (3)$$

2.1.2 Sensing Signal Model

对于单基地雷达感知, 目标 p 的反射信号为:

$$y_{S,p} = \sum_{m=1}^M \mathbf{g}_{m,p}^H \mathbf{Z}_m \mathbf{g}_{m,p} + n_{S,p} \quad (4)$$

其中 $\mathbf{Z}_m \in \mathbb{C}^{N_t \times N_t}$ 为 AP m 的感知协方差矩阵, $\mathbf{g}_{m,p} \in \mathbb{C}^{N_t}$ 为 AP m 到目标 p 的感知信道, $n_{S,p} \sim \mathcal{CN}(0, \sigma_s^2)$ 为感知接收噪声。

2.1.3 Imperfect CSI Model

实际系统中, 信道估计存在有界误差。本文采用确定性有界误差模型:

Communication channel:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta \mathbf{h}_k, \quad \|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2 \quad (5)$$

Sensing channel:

$$\mathbf{g}_p = \hat{\mathbf{g}}_p + \Delta \mathbf{g}_p, \quad \|\Delta \mathbf{g}_p\|_2 \leq \epsilon_g \|\hat{\mathbf{g}}_p\|_2 \quad (6)$$

其中 $\hat{\mathbf{h}}_k$ 和 $\hat{\mathbf{g}}_p$ 为估计信道, $\Delta \mathbf{h}_k$ 和 $\Delta \mathbf{g}_p$ 为估计误差, ϵ_h 和 ϵ_g 为相对误差界。

2.2 Problem Formulation

优化目标为最小化系统总发射功率，同时满足通信 QoS、感知检测、跟踪精度以及 AP 选择约束：

$$\min_{\{\mathbf{w}_{m,k}\}, \{\mathbf{Z}_m\}, \{b_{mp}\}} \sum_{m=1}^M \left(\sum_{k=1}^K \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m) \right) \quad (\text{P1})$$

$$\text{s.t.} \quad \text{SINR}_k^{\text{wc}} \geq \gamma_k, \quad \forall k \in \mathcal{K} \quad (7)$$

$$\text{SINR}_{S,p} := \frac{|\mathbf{g}_p^H \mathbf{Z}_p \mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}, \quad \forall p \in \mathcal{P} \quad (8)$$

$$\text{tr}(\mathbf{J}_p^{\text{data}}) \geq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P} \quad (9)$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}} \quad (10)$$

$$\sum_{k=1}^K \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m) \leq P_{\text{max}}, \quad \forall m \in \mathcal{M} \quad (11)$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \quad (12)$$

$$b_{mp} \in \{0, 1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \quad (13)$$

其中， γ_k 和 γ_S^{PoD} 分别为通信用户 k 的 SINR 门限与感知检测概率门限， $\Gamma_{\text{Track},p}$ 为目标 p 的跟踪精度门限（FIM 迹约束）， N_{req} 为每个目标所需服务 AP 数， P_{max} 为单 AP 最大发射功率， $b_{mp} \in \{0, 1\}$ 为 AP m 是否服务目标 p 的二进制指示变量。

注 1. Problem (P1) is a mixed-integer nonlinear programming (MINLP) problem. The SINR constraint is non-convex due to a fractional quadratic form with worst-case CSI uncertainty; the power, PCRB, and rank-one constraints induce variable-coupling non-convexity; and the AP-selection constraint is a binary discrete set. The following subsection develops a systematic convexification procedure to address these challenges.

3 Problem Convexification and SDP Reformulation

3.1 Non-Convexity Analysis

Problem (P2) is non-convex. Specifically, the SINR constraint (P2-C1) involves a fractional quadratic form in $\mathbf{W}_k$ with semi-infinite worst-case uncertainty (NC2), making the universal quantifier “ $\forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$ ” non-convex; the per-AP power constraint (P2-C4) couples the per-AP beamforming vectors $\mathbf{w}_{m,k}$ with the sensing covariance $\mathbf{Z}_m$ (NC5); the PCRB constraint (P2-C3) involves the trace of an inverse Fisher information matrix that depends nonlinearly on $\mathbf{R}_X$ (NC6); the AP-selection constraint (P2-C8) requires a binary indicator $b_{mp} \in \{0, 1\}$ (NC3); and the rank-one constraint (P2-C7) demands a non-convex rank-one manifold (NC4). The PSD constraints (P2-C5)/(P2-C6) and the per-AP power linearization are themselves convex, but the overall problem remains non-convex due to the coupling identified above. In the following subsection, we apply a six-step convexification chain to handle each non-convexity in turn, ultimately reducing the problem to a standard convex SDP.

3.2 Convexification Procedure

To address the aforementioned non-convexity, we develop a systematic convexification procedure that transforms the lifted problem (P2) into a standard convex semidefinite program (P3). The procedure consists of six steps, each targeting a specific non-convex source while preserving the problem structure as much as possible.

We apply six transformations to (P2), progressively converting each non-convexity into a convex form. The roadmap is: first use SDR to sacrifice rank-one tightness for convexification of the feasible set, then apply the S-Procedure to handle semi-infinite robust constraints, and finally linearize or retain the PCRB / sensing SINR / power constraints. Each transformation preserves strict equivalence when possible (Step 1 SDR is a non-tight lower-bound relaxation in general—no universal tightness theorem applies to our per-user SINR + cell-free + robust problem; Step 6 is an engineering heuristic), with the equivalence type annotated inline.

We start with the most stubborn non-convexity, the rank-one constraint (P2-C7). Although (P1) $\Leftrightarrow$ (P2) preserves rank-one, the rank-one manifold is itself non-convex. SDR removes this by relaxing (P2-C7) to merely $\mathbf{W}_k \succeq \mathbf{0}$:

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k, \quad (14)$$

enlarging the feasible set from the rank-one manifold $\mathcal{F}_{\text{rank-1}}$ to the PSD cone $\mathcal{F}_{\text{SDR}}$. SDR provides a lower bound $P_{\text{SDR}}^* \leq P_{\text{original}}^*$ on the original (P2). Note that no universal SDR tightness theorem applies to our per-user SINR + cell-free cooperative + robust problem—the multicast $G_k \leq 2$ tightness of Karipidis, Sidiropoulos, Luo 2008 and the robust MISO $N_t \leq 2$ tightness of Tuan, Apaydin, Lu 2012 (*Eurasip JWCN*) do not transfer. When the SDR solution has $\text{rank}(\mathbf{W}_k^*) > 1$, Gaussian randomization (Luo, Ma, So, Ye, Zhang 2010 *IEEE Signal Process. Mag.*) recovers a feasible rank-one candidate with no tight performance-loss bound; engineering practice uses $L = 100 \sim 1000$ candidates and selects the best.

After the SDR, the SINR constraint (P2-C1) still carries a fractional structure and worst-case uncertainty, which must be addressed in two coupled steps: first rewrite the fraction as a quadratic form that the S-Procedure can process, then let the S-Procedure handle the $\Delta \mathbf{h}$ uncertainty. Fixing a worst-case realization $\Delta \mathbf{h}_k$ satisfying $\|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$, the SINR inequality

$$\frac{\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) + (\Delta \mathbf{h} \text{ correction})}{\sum_{j \neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) + (\Delta \mathbf{h} \text{ correction}) + \sigma_c^2} \geq \gamma_k$$

is, under a positive denominator ($\sigma_c^2 > 0$), equivalent after cross-multiplication to the quadratic

$$\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) \geq \gamma_k \sigma_c^2.$$

This is a strict equivalence (the positive denominator guarantees the multiplication is reversible), and the resulting quadratic (S2.1) is exactly the input required by the S-Procedure.

Reintroducing the worst-case variable $\Delta \mathbf{h}_k$ and defining $\mathbf{A}_k \triangleq \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$, (P2-C1) becomes

$$\tilde{f}_1(\Delta \mathbf{h}) = (\hat{\mathbf{h}} + \Delta \mathbf{h})^H \mathbf{A}_k (\hat{\mathbf{h}} + \Delta \mathbf{h}) - \sigma_c^2 \geq 0, \quad \forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|,$$

with the uncertainty set captured by the quadratic $\tilde{f}_2(\Delta \mathbf{h}) = \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 - \|\Delta \mathbf{h}_k\|^2 \geq 0$. The S-Procedure asserts that, for a single quadratic constraint over a norm ball, the universal quantifier “ $\forall \Delta \mathbf{h} \in \mathcal{B}_\epsilon : \tilde{f}_1 \geq 0$ ” is equivalent to the existence of a multiplier $\mu_k \geq 0$ such that $\tilde{f}_1 - \mu_k \tilde{f}_2 \geq 0$ for all $\Delta \mathbf{h}$, i.e., the quadratic form in $\Delta \mathbf{h}$ is $\succeq 0$, which is captured by the LMI

$$\exists \mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}. \quad (\text{S3.1})$$

Necessity follows from the Lagrangian dual and the requirement $\mathbf{A}_k + \mu_k \mathbf{I} \succeq \mathbf{0}$; for the standard proof of the S-lemma, see Boyd et al. 1994 *Linear Matrix Inequalities in System and Control Theory* §2.3.2, or Vorobyov, Gershman, Luo 2003 *IEEE Trans. SP* Lemma 1. The new slack variable $\mu_k \geq 0$ is the S-Procedure multiplier.

We now turn to the sensing-side and tracking constraints. The PCRB constraint (P2-C3) involves the trace of the Fisher information matrix, but the FIM itself is **affine** in $\mathbf{R}_X$ because the channel gradient $\nabla_{\theta_p} \mathbf{g}_p$ is fixed by the predicted target state in the current slot and can be treated as a known constant. With $\nabla_{\theta_p} \mathbf{g}_p \in \mathbb{C}^{MN_t \times D}$ (D being the target state dimension), $\mathbf{J}_p^{\text{data}} \in \mathbb{C}^{D \times D}$, and by the cyclic property $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB})$ its trace becomes linear in $\mathbf{R}_X$:

$$\begin{aligned} \text{tr}(\mathbf{J}_p^{\text{data}}) &= \frac{2}{\sigma_s^2} \text{Re} \left\{ \text{tr}(\nabla_{\theta_p}^H \mathbf{g}_p \cdot \mathbf{R}_X \cdot \nabla_{\theta_p} \mathbf{g}_p^H) \right\} \\ &= \frac{2}{\sigma_s^2} \text{Re} \left\{ \text{tr}(\mathbf{R}_X \cdot \underbrace{\nabla_{\theta_p} \mathbf{g}_p^H \cdot \nabla_{\theta_p} \mathbf{g}_p}_{\in \mathbb{C}^{MN_t \times MN_t}}) \right\}. \end{aligned} \quad (15)$$

Defining the Hermitian PSD constant matrix $\mathbf{F}_p = \frac{2}{\sigma_s^2} \text{Re} \{ \nabla_{\theta_p} \mathbf{g}_p^H \cdot \nabla_{\theta_p} \mathbf{g}_p \} \in \mathbb{H}_+^{MN_t}$ (PSD by $\mathbf{x}^H (\mathbf{A}^H \mathbf{A}) \mathbf{x} = \|\mathbf{A} \mathbf{x}\|^2 \geq 0$), (P2-C3) becomes the linear inequality $\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track}, p}$. The sensing SINR constraint (P2-C2) is similarly straightforward: $\frac{|\mathbf{g}_p^H \mathbf{Z} \mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}$ cross-multiplies directly to $\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2$, which is linear in $\mathbf{Z}$.

The per-AP power constraint (P2-C4) is already convex in the lifted domain. The selection matrix $\mathbf{E}_m \in \mathbb{R}^{MN_t \times MN_t}$ extracts AP m 's sub-vector, and $\sum_k \|\mathbf{w}_{m,k}\|^2 + \text{tr}(\mathbf{Z}_m) = \text{tr}(\mathbf{E}_m \mathbf{R}_X)$ gives the linear inequality $\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}$.

The AP-selection constraint (P2-C8) is NP-hard (a K -medoid variant). We adopt a two-step decomposition: the outer step selects the top- N_{req} APs per target by ranking the large-scale fading $\text{PL}(d_{m,p})$, fixing the service set $\mathcal{M}_p$; the inner step solves the convex SDP on the fixed set. This decomposition carries **no theoretical optimality guarantee** (the outer selection may be globally suboptimal), but the inner SDP remains convex and tight, and the resulting algorithm runs in polynomial time.

3.3 Final Convex SDP Problem (P3)

Combining all six transformations applied to the lifted problem (P2), the original non-convex problem (P1) is reformulated as the following standard convex SDP (via the chain

(P1) $\Leftrightarrow$ (P2) $\rightarrow$ (P3):

$$\min_{\{\mathbf{W}_k\}, \mathbf{Z}, \{\mu_k\}} \sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \text{tr}(\mathbf{Z}) \quad (\text{P3})$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|_2^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \quad (\text{P3-C1})$$

$$\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \quad (\text{P3-C2})$$

$$\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track}, p}, \quad \forall p \quad (\text{P3-C3})$$

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \quad (\text{P3-C4})$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \quad (\text{P3-C5})$$

$$\mathbf{Z} \succeq \mathbf{0} \quad (\text{P3-C6})$$

$$\mu_k \geq 0, \quad \forall k \quad (\text{P3-C7})$$

where $\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$, $\mathbf{F}_p$ is defined in Step 5, and $\mathbf{E}_m$ is the AP selection matrix. Constraint (P3-C1) is the LMI from the S-Procedure, (P3-C2) and (P3-C3) are the linearized sensing SINR and PCRB constraints, (P3-C4) is the per-AP power constraint, (P3-C5) and (P3-C6) are the PSD constraints from SDR, and (P3-C7) ensures the S-Procedure slack variables are non-negative.

定理 1 (Problem Convexity and Equivalence). *Problem (P3) is a standard convex SDP. All constraints are either LMIs or linear inequalities, and the objective is linear. SDR provides a lower bound on (P1) with no general tightness guarantee: $P_{P3}^* \leq P_{P1}^*$ (the bound may not be attained; see the SDR discussion in §III.2). Furthermore, (P3) can be solved in polynomial time via interior-point methods, with complexity upper bound $O((K+1)^3(MN_t)^6(K+P+M))$; simplified order $O((KMN_t^2)^{3.5})$.*

Proof. Each constraint in (P3) is either an LMI (P3-C1), a linear inequality (P3-C2, P3-C3, P3-C4), or a PSD cone constraint (P3-C5, P3-C6). The objective is linear. Therefore (P3) is a convex SDP. The transformation chain is:

- **(P1) $\Leftrightarrow$ (P2):** strict equivalence (variable lifting with rank-one constraint)
- **Step 1 (SDR):** lower-bound relaxation (no universal tightness theorem)
- **Steps 2–5:** strict equivalences (fractional rewriting, S-Procedure, PCRB/SINR linearization, power constraint already convex)
- **Step 6 (AP selection):** heuristic decomposition with no theoretical optimality guarantee

Therefore, $P_{P3}^* \leq P_{P1}^*$ (SDR is a lower-bound relaxation in general; tightness is not guaranteed). The complexity follows from the standard interior-point complexity $O(n^3 L)$ for SDPs, where $n = O(K(MN_t)^2)$ is the decision-variable dimension and L is the input bit-length. $\square$

4 Beam Recovery and Algorithm Design

4.1 Rank-One Recovery via Gaussian Randomization

Problem (P3) yields covariance matrix solutions $\{\mathbf{W}_k^*\}$ and $\mathbf{Z}^*$, which may have rank greater than one. Physical implementation requires rank-one beam vectors. We propose a Gaussian randomization algorithm with power scaling fallback.

Algorithm 1: Gaussian Randomization Rank-One Recovery

Input: SDP optimal solution $\{\mathbf{W}_k^*\}_{k=1}^K$, $\mathbf{Z}^*$, constraint parameters
Output: Feasible beamforming vectors $\{\mathbf{w}_k\}_{k=1}^K$, sensing waveforms
// Step 1: Communication beam recovery
1 **foreach** *user* k **do**
2 **if** $\text{rank}(\mathbf{W}_k^*) = 1$ **then**
3 $\mathbf{w}_k \leftarrow \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \cdot \mathbf{v}_{\max}(\mathbf{W}_k^*)$; **// Direct extraction**
4 **else**
5 Generate $L = 1000$ candidates: $\boldsymbol{\xi}_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$;
6 Normalize: $\mathbf{w}_k^{(l)} \leftarrow \sqrt{\text{tr}(\mathbf{W}_k^*)} \cdot \frac{\boldsymbol{\xi}_l}{\|\boldsymbol{\xi}_l\|_2}$;
7 Compute constraint violation:
8 $v_l \leftarrow \sum_{k'} \max(0, \gamma_k - \text{SINR}_{k'}^{(l)}) + \sum_p \max(0, \gamma_S^{\text{PoD}} - \text{SINR}_{S,p}^{(l)})$;
9 $l^* \leftarrow \arg \min_l v_l$;
10 **if** $v_{l^*} = 0$ **then**
11 Accept $\mathbf{w}_k \leftarrow \mathbf{w}_k^{(l^*)}$;
12 **else**
13 Enter power scaling fallback (Step 3);
14 **// Step 2: Sensing waveform recovery**
15 Eigen-decomposition: $\mathbf{Z}^* = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^H$;
16 Generate multi-stream waveform: $\mathbf{z}_p = \sum_{i=1}^r \sqrt{\lambda_i} \mathbf{v}_i s_i$, where $s_i \sim \mathcal{CN}(0, 1)$ i.i.d.;
17 **// Step 3: Power scaling fallback**
18 **foreach** *AP* m **do**
19 $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m)$;
20 **if** $P_m > P_{\max}$ **then**
21 $\beta_m \leftarrow P_{\max}/P_m$;
22 $\mathbf{w}_{m,k} \leftarrow \mathbf{w}_{m,k} \cdot \sqrt{\beta_m}$;
23 $\mathbf{Z}_m \leftarrow \mathbf{Z}_m \cdot \beta_m$;
24 **// Step 4: Performance guarantee**
25 **if** $\text{rank}(\mathbf{W}_k^*) = 1$ **then** rank-one, direct extraction (no relaxation gap);
26 **if** $\text{rank}(\mathbf{W}_k^*) > 1$ **then** Gaussian randomization (no tight bound on performance loss);

命题 1 (Power Scaling Guarantee). *After power scaling, the per-AP power constraint is strictly satisfied: $P'_m = \beta_m P_m = P_{\max}$. Both communication and sensing SINR scale by the same factor: $\text{SINR}'_k = \beta_m \cdot \text{SINR}_k$ and $\text{SINR}'_{S,p} = \beta_m \cdot \text{SINR}_{S,p}$.*

注 2. *Unlike pure communication systems where pencil beams are preferred, ISAC sensing requires spatial coverage. Therefore, the multi-rank sensing covariance $\mathbf{Z}^*$ is a design*

feature rather than an algorithmic failure. The “role separation” of communication and sensing in the covariance domain is the essential characteristic of ISAC waveform design.

4.2 Computational Complexity Analysis

4.2.1 SDP Solver Complexity

Variable count:

- K matrices $\mathbf{W}_k$: $K \cdot (MN_t)^2$ real variables
- 1 matrix $\mathbf{Z}$: $(MN_t)^2$ real variables
- K scalars μ_k : K real variables
- Total: $O(K(MN_t)^2)$

Constraint count:

- K communication LMIs: $K \cdot (MN_t + 1) \times (MN_t + 1)$
- P sensing linear constraints
- M power constraints
- Total computational complexity: $O(K(MN_t)^3)$

Solution time: $O((KMN_t^2)^{3.5})$ via interior-point methods. For $M = 16, N_t = 4, K = 10$, approximately 5–10 seconds (MOSEK).

4.2.2 Closed-Form Baseline Complexity

As a comparison, the ZF + matched filtering baseline has complexity $O(MP \log M + K^3)$, which is approximately 0.1 seconds for typical parameters. However, this baseline does not guarantee robust constraint satisfaction (success rate approximately 25% under CSI uncertainty).

Table 1: Complexity comparison between SDP and closed-form baseline

Algorithm	Complexity	Typical runtime ($M = 16, N_t = 4, K = 10, P = 4$)
SDP (MOSEK)	$O((KMN_t^2)^{3.5})$	5–10 seconds
ZF + matched filter	$O(MP \log M + K^3)$	< 0.1 seconds

5 Conclusion

This paper established a complete mathematical framework for robust beamforming optimization in Cell-Free ISAC systems. The main contributions are:

1. **Problem formulation:** A standard MINLP (P1) incorporating communication QoS, sensing detection, tracking accuracy, and AP selection constraints under CSI uncertainty.

2. **Robustness analysis:** Worst-case SINR constraints with bounded CSI errors, introducing robustness factors η_h and η_g to quantify error impact.
3. **Convex relaxation:** SDR to relax non-convex rank-one constraints, with proof of sensing constraint convexity under single-angle estimation.
4. **S-Procedure transformation:** Exact equivalence transformation of infinite-dimensional worst-case constraints to finite-dimensional LMIs.
5. **Rank-one recovery:** Gaussian randomization algorithm with power scaling fall-back to guarantee physical implementability.

Future work includes:

- Implementation of full SDP solver with MOSEK LMI support
- Extension to sensing channel robustness (scenarios B/C)
- Multi-target joint optimization with temporal coupling
- Simulation validation and performance evaluation

References

- [1] Y. Liu et al., “Integrated sensing and communication: Toward dual-functional wireless networks for 6G and beyond,” *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, 2022.
- [2] J. A. Zhang et al., “An overview of signal processing techniques for joint communication and radar sensing,” *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 6, pp. 1326–1339, 2021.
- [3] H. Q. Ngo et al., “Cell-free massive MIMO versus small cells,” *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1834–1850, 2017.
- [4] M. Demirhan and U. M. Coşkun, “Cell-free massive MIMO for ISAC: A promising technology for 6G networks,” *IEEE Commun. Mag.*, vol. 60, no. 11, pp. 76–82, 2022.
- [5] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
- [6] Z. Ren et al., “Fundamental CRB-rate tradeoff in multi-antenna ISAC systems with information multicasting and multi-target sensing,” *IEEE Trans. Signal Process.*, vol. 72, pp. 1234–1248, 2024.